Package ‘GenotypeQuant’

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Type Package

Title Estimates the number of genotypes present in a eDNA sample genotyped with microsatellite markers

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Author Filipe Alberto & Lauran Liggan

Maintainer Filipe Alberto <albertof@uwm.edu>

Description The method uses the count of unique (different) microsatellite alleles amplified from an eDNA sample
to estimate the number of genotypes present in the sample using a numerical maximum likelihood approach.
In other words, the estimate is the number of genotypes that most likely generated the observed counts of different alleles for each locus. The method relies on resampling an allele frequency table for the reference population
to generate sampling distribution for the counts of unique alleles expected in samples containing different numbers of genes
(haploid genotypes). Users can estimate the number of haploid or haploid genotypes. A correction for FIS is available.

Imports adegenet, genepop

NeedsCompilation no

R topics documented:

G.Estimate — *Estimates the most likely number of haploid or diploid genotypes in an eDNA sample using the observed counts of unique microsatellite alleles amplified for a set of markers.*

Description

The function compares a user supplied vector with counts of unique alleles observed (genotyped) for each locus with sampling distributions created from a reference population MakeAllNumberDist. A numerical maximum likelihood method is used to estimate the most likely number of haploid or diploid genotypes that might have generated the observed counts of unique alleles observed across loci.

Usage

```
G.Estimate(Obs.N.all,All.Num.Dist,ML.curve=TRUE,ploidy=1,
col="black",col.est="darkcyan",col.ci="darkcyan")
```

Arguments

`Obs.N.all`	A vector with length equal to the number of locus in the reference population allele frequency file. The values are the counts of unique alleles genotyped at each locus from an eDNA sample. Locus with missing data should be coded with NA.
`All.Num.Dist`	The 3-D array produced by MakeAllNumberDist. This array contains the sample distributions with unique allele counts generated by sampling the reference population allele frequencies, with or without Fis correction.
`ML.curve`	A logical to decide if a plot with the maximum-likelihood curve should be produced; defaults to TRUE.
`ploidy`	If estimating the number of haploids genotypes set to 1, if diploids set to 2.
`col`	The color of the ML-curve line.
`col.est`	The color of the vertical line intersecting the ML-curve at the estimated number of gametophytes.
`col.ci`	The color of the vertical lines intersecting the ML-curve at the 95 percent confidence interval.

Details

The probability *P(OCUA | g)* that a given count of unique microstallite alleles *OCUA* (the vector elements in function argument **Obs.N.all**) is amplified at locus *i* from an eDNA sample was produced by *g* genotypes is estimated using a numerical maximum likelihod approach. The relative frequency of *OCUA* in different sampling distributions of expected counts of unique allele (ECUA) is calculated for samples of size *g*, from 1 to *G*. The multilocus *P(OCUA | g)* is calculated by summing the log probabilities for all loci for each *g* value. The maximum likelihood estimate is simply the maximum multilocus probability across the values of *g*. The sampling distributions for each locus and *g* combination are created by sampling with replacement samples of size *g* alleles according to their frequencies in the reference population, with or without Fis correction, and counting the number of unique alleles *CUA* in each sample. The sampled values of *CUA*, building each sampling distribution, are recorded in **All.Num.Dist**. An approximate confidence interval is produced using the likelihood ratio approach; in brief, the 95 percent interval includes all genotype values for which the log-likelihood function drops off by no more than 1.92 units. If estimating diploids the function simply assumes that the number of diploid genotypes is the double of haploid ones. Thus, for example, the distribution for four diploid genotypes is the same as the one generated for samples of eight genes. An eDNA sample might fail to amplify all the microsatellite alleles present. More concerning, for problematic samples, there might be heterogeneity across loci in the number of alleles that are amplified. This can result in observed allele counts across loci that do not follow the different allele richness of each loci in the reference population. For example, a marker with more alleles in the reference population might have fewer amplified alleles in the eDNA sample than a loci with fewer alleles in the reference population. This will result in a biased low estimate of the number of genotypes present and wide, non-informative confidence intervals. We used a simple chi-square test to measure how the observed number of alleles compared with the expected number of alleles, given the diversity of the different loci in the reference population and the total number of alleles amplified. The chi-square test measures the quality of the eDNA sample. It provides complementary information to the CI because a lower allele richness, number of loci used, or both can also result in wide CI for eDNA samples with high quality (i.e., without amplification problems).

Value

A list with the following components:

`estimateM`	A vector with three elements, the most likely number of genotypes (G.hat), and the lower and upper bounds of the 95 percent confidence interval.
`LogML`	The log-likelihoods for all values of genotypes from one to the maximum number available in the sampling distributions. Inf values are returned when *n.obs.A* are impossible given a particular number of genotypes.
`Amp.Qual`	The results from a Chi-square test comparing the number of observed alleles scored in an eDNA sample with the expected number of alleles given loci diversity in the reference population and the total number of alleles scored. The statistic ((0-E)^2)/E is given for each locus to allow detection of which loci deviate more from expectations; the sum of these values is given, the Chi-square statistic, as well as the p-value wi degrees of freedom equal to the number of loci minus two.

Author(s)

F. Alberto and L. Liggan.

References

Liggan L, Alberto F (in prep)...

See Also

MakeAllNumberDist

`MakeAllNumberDist` *Generates distributions for the counts of unique alleles in samples with different size g (number of haploid genotypes).*

Description

By sampling with replacement from the allele frequency table for the reference population, with or without Fis correction, this function generates sampling distributions for the counts of unique alleles for samples of different size g (number of haploid genotypes).

Usage

```
MakeAllNumberDist(genotypes.df,ngametos=80,All.nchar=3,Fis=FALSE,Fis.treshold=0.1,
npermutations=5000)
```

Arguments

`genotypes.df`	The input file with the reference population microsatellite allele codes. This is a data.frame with one column per locus one row per individual with column names containing the locus names and without individual information (i.e. the first column is the information for the first loci). Use the same number of digits to code each allele. For example, 120126 is a possible value for a diploid coded with three digits per allele. At this stage, the reference data needs to be in diploid format. Missing data needs to be coded with 0, 00, 0000, or 000000..
`ngametos`	The method will generate sampling distributions for the counts of unique alleles for samples of different size g (number of haploid genotypes). Distributions are sampled from the reference allele frequency table. There are distributions for each combination of locus and number of genotypes from 1 to *ngametos*.
`All.nchar`	How many digits are used in the reference population file to code each allele (can only be 1, 2, or 3).
`Fis`	A logical indicating if Fis corrections should be applied when generating the sampling distributions. If yes, Genepop is used to calculate Qintra, the observed frequencies of identical pairs of genes in the reference population. Alleles are then sampled sequential with probability Qintra to sample the previously sampled allele and 1-Qintra to sample any other allele according to their allelic frequencies. Defaults to FALSE.
`Fis.treshold`	When applying Fis correction, it is recommended to only correct for Fis values above a certain threshold. The default is 0.1.
`npermutations`	How many samples should each sampling distribution contain; defaults to 5000.

Details

The probability *P(OCUA | g)* that a given count of unique microstallite alleles *OCUA* is amplified at locus i from an eDNA sample was produced by g genotypes is estimated using a numerical maximum likelihod approach. The relative frequency of *OCUA* in different sampling distributions of expected counts of unique allele (ECUA) is calculated for samples of size g, from 1 to G. The multilocus *P(OCUA | g)* is calculated by summing the log probabilities for all loci for each g value. The maximum likelihood estimate is simply the maximum multilocus probability across the values of g. The sampling distributions for each locus and g combination are created by sampling with replacement samples of size g alleles according to their frequencies in the reference population, with or without Fis correction, and counting the number of unique alleles *CUA* in each sample.

Value

A three-dimensional array with dimensions ngametos, number of loci (in the reference population allele frequency table), and npermutations, containing values for counts of unique alleles sampled. This object will be required by the function `G.Estimate` to estimate the most likely number of genotypes present in the eDNA sample.

Author(s)

F. Alberto and L. Liggan.

References

Liggan L, Alberto F (in prep)...

See Also

`G.Estimate`

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